LAKSHANA S

Electronics & Communication Engineering Student | VLSI Design & Embedded Systems Enthusiast

+91 - 99526 06191 — lakshanamani70@gmail.com — Karur, Tamil Nadu - 639004

PROFILE SUMMARY

Motivated and enthusiastic Electronics and Communication Engineering student with a strong foundation in VLSI Design, Embedded Systems, IoT, and Digital Electronics. Passionate about designing innovative hardware solutions and developing efficient embedded applications, with hands-on project and internship experience. Eager to apply technical knowledge, problem-solving skills, and research experience to contribute to a challenging organization while continuously learning.

AREA OF INTEREST

• VLSI Design • Embedded Systems • IoT Applications

INTERNSHIP

VLSI Design and Verification Intern — Taras Training and Solutions (15 Days)

- Gained practical knowledge of the VLSI design and verification flow.
- Learned Digital Electronics, Verilog HDL, and SystemVerilog fundamentals.
- Developed and simulated RTL designs using industry-standard design methodologies.
- Performed functional verification, debugging, and waveform analysis.

ACADEMIC PROJECTS

Malnutrition Detection in Children

- Developed a system for early detection of malnutrition in children.
- Analyzed health parameters to identify nutritional deficiencies and improve healthcare monitoring.

IoT-Based Smart Health Monitoring

- Designed a real-time health monitoring system using Arduino Uno.
- Integrated a temperature sensor, LCD display, and buzzer for patient monitoring with automated alerts.

Performance Analysis of Memristor-Based Memory Circuits

- Simulated and analyzed the performance of memristor-based memory circuits.
- Evaluated switching behavior, power consumption, and memory retention against conventional memory technologies.

SKILLS

Programming Languages: C (Basic), Python (Basic)
Hardware Description Languages: Verilog HDL, SystemVerilog
VLSI Design: Digital Design
Embedded Systems: Arduino UNO, ESP32
IoT: Arduino-based IoT Applications, Smart Agriculture Monitoring
Tools & Simulation: Wokwi, TinkerCAD, Arduino IDE
Core Subjects: Digital Electronics, Embedded Systems, VLSI, Communication Systems
Soft Skills: Communication, Teamwork, Self-Motivation, Active Listening, Leadership
Languages Known: Tamil, English, Kannada

EDUCATION

Qualification	School / College	Year	Board / University	Percentage
B.E. ECE	Chettinad College of Engineering and Technology, Karur, Tamil Nadu	Pursuing	Anna University	8.06% (up to 5th Sem)
HSC	Bharani Park Matric Higher Secondary School	2023	State Board	75.6%

ACHIEVEMENTS

- **Paper Presentation, First Prize** — Selvam Engineering College, for "MXene: Advanced Two-Dimensional Nanomaterials for Next-Generation Applications" (Feb 2025)
- **Paper Presentation** — Participated at Kongunadu College of Engineering and Technology on Brain-Computer Interface (BCI) (Aug 2026)

REFERENCE

Dr. G. Sahaya Dennish Babu, Associate Professor of Physics, Chettinad College of Engineering & Technology, Karur, TN.
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